Software Test Plan

Web-Based ML Model Evaluator

Complete document: Sections 1 to 15 and Appendix A

Project	Web-Based ML Model Evaluator
Repository	SE_498_515_534_911
Version	1.0
Authors	Members 1, 2, 3 and 4
Date	13-10-2026
Status	Integrated, for team review
Related documents	SRS (Members 1 and 2), SAD (Members 3 and 4), RTM

Revision History

Version	Date	Author	Change summary
0.1	10-10-2026	Member 1	Initial draft of all sections, including 10 test cases and the traceability matrix
0.2	12-10-2026	Members 1, 2, 3, 4	Integrated sections; aligned requirement IDs and security traceability
1.0	13-10-2026	Members 1, 2, 3, 4	Sections 10, 11, 12, 14 and 15 restored; Appendix A replaced by Member 4's ten test cases; Section 13 extended with design references from SAD Appendix A; SAD cross-references updated

Contributions

Member	Sections
Member 1	1 Introduction, 2 Test Items, 6 Test Environment, 7 Test Schedule, 8 Test Deliverables, 9 Roles; final merge and formatting
Member 2	3 Features to be Tested, 4 Features not to be Tested, 5 Test Approach, 5.1 Security Validation, 13 Traceability (requirement side)
Member 3	10 Risks, 11 Assumptions, 12 Suspension and Resumption, 14 Metrics, 15 Approvals
Member 4	Appendix A (seven functional and three non-functional test cases) and the design-reference column of Section 13

1. Introduction

Purpose: This document defines the test plan for the Web-Based ML Model Evaluator (WBMME) v1.0. It describes the objectives, scope, strategy, resources, schedule and responsibilities for testing, and lists the test cases that verify the requirements in the SRS.

Scope: Testing covers user registration and login, role-based access, CSV dataset upload and preparation, model selection and training, evaluation metrics and charts, experiment history and comparison, report export, administration, and the stated performance, portability, usability and security requirements. Internals of third-party libraries and the hosting platform are excluded.

References: WBMME SRS v1.0; WBMME SAD v1.0 (complete document, Sections 1 to 5); SE Mini-Project Deliverables Part-1 and Part-2; Jackfruit SE Mini-Project Guidelines 2026; IEEE 829 (test documentation); OWASP Top 10.

Definitions: WBMME (Web-Based ML Model Evaluator), SRS (Software Requirements Specification), SAD (Software Architecture Document), RTM (Requirements Traceability Matrix), TC (Test Case), CSV (Comma-Separated Values), UAT (User Acceptance Testing), CSRF (Cross-Site Request Forgery), STRIDE (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege).

2. Test Items

- Authentication and access control module (registration, login, roles)
- Data workspace module (CSV upload, validation, profiling, preprocessing, train/test split)
- Model selection and training module
- Evaluation module (metrics, confusion matrix, feature importance)
- Experiment management and reporting module (history, comparison, export)
- Admin interface (user and dataset management)
- REST/JSON endpoints: /api/upload, /api/train, /api/evaluate, /api/runs
- Logging, monitoring and input validation controls

3. Features to be Tested

Features are listed with the requirement IDs of the SRS (Part B, sections 4 and 5). “Covered by” names the test case (Appendix A) or the security check (section 5.1) that verifies the requirement; “planned” means that it is scheduled in the sprint shown and has no test case among the ten in Appendix A yet.

3.1 Functional features

Req ID	Feature	Priority	Covered by
MLE-F-001	User registration and login	High	TC-AUTH-01
MLE-F-002	Role-based access (Data Scientist, Student, Admin)	High	TC-SEC-01

Req ID	Feature	Priority	Covered by
MLE-F-003	Upload CSV dataset with validation	High	TC-DATA-01, TC-DATA-02
MLE-F-004	Preview and profile the dataset	High	TC-DATA-01
MLE-F-005	Handle missing values and encode categories	High	Planned (Sprint 1)
MLE-F-006	Configure target column and train/test split	High	TC-MODEL-01
MLE-F-007	Select a model from the catalogue	High	TC-MODEL-01
MLE-F-008	Configure hyperparameters	Medium	TC-MODEL-01
MLE-F-009	Train a model and show status	High	TC-MODEL-01
MLE-F-010	Compute accuracy, precision, recall, F1	High	TC-EVAL-01
MLE-F-011	Show confusion matrix and plots	High	TC-EVAL-01
MLE-F-012	Feature importance / explanation	Medium	Planned (Sprint 2)
MLE-F-013	Save each run (history)	High	TC-EXP-01
MLE-F-014	Compare runs side by side	Medium	TC-EXP-01
MLE-F-015	Export results report	Medium	TC-DATA-02 (step 3) (planned, Sprint 2)
MLE-F-016	Admin manages users and datasets	High	TC-SEC-01

3.2 Non-functional features

Req ID	Feature	Priority	Covered by
MLE-NF-001	Upload of up to 10 MB processed in 5 s or less	High	TC-PERF-01
MLE-NF-002	Training of standard models on up to 50,000 rows in 60 s or less	High	TC-PERF-02
MLE-NF-003	Pages load in 3 s or less for 20 concurrent users	Medium	TC-PERF-03
MLE-NF-004	Runs on Windows, Linux and macOS via Docker	Medium	Planned (Sprint 2)
MLE-NF-005	Usable by a first-time user without training	Medium	Planned (Sprint 2)
MLE-NF-006 (new)	Failed jobs are contained; saved data survives restart	Medium	Planned (Sprint 2)
MLE-NF-007 (new)	Keeps page-load limits with 20 datasets and 100 runs per user	Low	Planned (Sprint 2)

3.3 Security features

Req ID	Feature	Priority	Covered by
MLE-SR-001	Passwords stored hashed	High	TC-AUTH-01, SV-01
MLE-SR-002	All traffic over HTTPS (TLS 1.2+)	High	SV-02 (planned, Sprint 2)
MLE-SR-003	Uploads validated for type, size and content	High	TC-DATA-02, SV-03
MLE-SR-004	Only system-created model artifacts are loaded	High	SV-04 (planned, Sprint 2)
MLE-SR-005	Per-user data isolation and audit log of actions	High	TC-SEC-01, SV-05
MLE-SR-006 (new)	Request, job and run-time limits against abuse	High	TC-MODEL-01 (job limit), SV-06 (planned, Sprint 2)
MLE-SR-007 (new)	Session expiry, login throttling and CSRF protection	High	TC-AUTH-01 (cookie flag only), SV-07, SV-08 (planned, Sprint 2)

4. Features not to be Tested

Item	Reason
Internal correctness of third-party libraries (Django, pandas, scikit-learn, joblib, Chart.js)	Assumed tested by their maintainers. WBMME is tested for how it uses them, for example by comparing metrics with a scikit-learn reference (TC-EVAL-01).
Predictive quality or production suitability of the models a user trains	The system only trains, evaluates and reports. It makes no claim about model quality (SRS 1.2).
Browser internals and operating-system behaviour	Out of the project's control. Only current Chrome and Firefox on Windows, Linux and macOS are supported (MLE-NF-004).
Mobile and tablet layouts	Outside the SRS operating environment (SRS 2.4).
SHAP explanations, PDF export and Celery job queues	Optional, not committed in the SRS (MLE-F-012, MLE-F-015, SAD 3.6). They are tested only if implemented.
Load beyond 20 concurrent users; production-scale deployment tuning	The SRS commits to 20 concurrent users (MLE-NF-003) and a Docker deployment only.
Penetration testing beyond the OWASP ZAP baseline scan and the abuse cases of section 5.1	Out of scope for a two-sprint student project; the checks in 5.1 cover each security requirement.
Hosting platform and network infrastructure	Not supplied by the project. TLS is checked on the deployed build only (SV-02).

5. Test Approach / Strategy

Levels

Level	Scope	Tools
Unit	Module-level tests of services: validation, preprocessing, training, metrics, access checks. Coverage target 70% or higher for service modules (section 14).	pytest, pytest-cov
Integration	API layer to services, services to database and file store, and the pipeline stages in sequence (Upload, Preprocess, Train, Evaluate, Record).	pytest, Django test client, Postman
System	End-to-end flows through the browser: login, upload, split, train, evaluate, history, compare, export, administration.	Selenium, manual execution
Acceptance (UAT)	Walkthrough of the demo scenarios AC-1 to AC-5 (SRS 6.2) by the team and the faculty reviewer.	Manual

Types

- **Functional testing** of every feature in section 3, using the test cases of Appendix A.
- **Regression testing** on every pull request, automated in GitHub Actions.
- **Performance testing** against MLE-NF-001, NF-002 and NF-003 on one fixed machine, with its specification recorded.
- **Usability testing** with five first-time users (MLE-NF-005).
- **Reliability testing** by fault injection (invalid data, time-out) and a server restart (MLE-NF-006).
- **Portability checks** by building and starting the Docker image on Windows, Linux and macOS (MLE-NF-004).
- **Security testing** as described in section 5.1.

Test design techniques

- **Boundary value analysis and equivalence partitioning** for every limit: file size (9.9, 10 and 10.1 MB), rows (19, 20, 50,000 and 50,001), parameter ranges (for example C = 0.01, 100, 0 and 101), test share (10% and 50%).
- **Valid and invalid data sets** for every input, as the course test-case template requires.
- **Oracle comparison:** metrics are compared with a scikit-learn reference script that uses the same split and seed (tolerance 0.001).
- **Error guessing and abuse cases** derived from the STRIDE table in SAD 3.9.

Defect severity

Severity	Meaning
Critical	Data loss, authentication bypass, or data exposed between users. Testing is suspended (section 12).
High	A committed requirement fails with no workaround, or a security check fails.
Medium	A requirement fails but a workaround exists, or a non-committed feature fails.
Low	Cosmetic or wording issues.

Entry criteria: A stable build is merged to the main branch, the test data and the test environment are ready, and the relevant SRS requirements are baselined.

Exit criteria: 100% of the planned test cases for the sprint are executed, no critical or high-severity defect remains open, all acceptance criteria (SRS 6.2) are satisfied, and deviations from the SRS, SAD and this plan are documented for the final demo.

5.1 Security Validation

Security validation verifies the three security objectives of SRS 5.1.1, which match the objectives of SAD 3.9: confidentiality (MLE-SO-001), integrity (MLE-SO-002) and availability (MLE-SO-003). Each check below verifies one security requirement and is traceable to it. The checks run on the deployed Docker build with HTTPS, which is available before Sprint 2 (section 11).

Table 5.1-1: Security objectives and their checks

Objective	Requirements	Checks	Threats (STRIDE)
Confidentiality MLE-SO-001	MLE-SR-001, SR-002, SR-005(a), SR-007; MLE-F-002	SV-01, SV-02, SV-05, SV-07, SV-08	Spoofing, information disclosure, elevation of privilege
Integrity MLE-SO-002	MLE-SR-003, SR-004, SR-005(b)	SV-03, SV-04, SV-05	Tampering, repudiation
Availability MLE-SO-003	MLE-SR-006	SV-06	Denial of service
All objectives	MLE-SR-001 to SR-007	SV-09, SV-10	All

Confidentiality checks

ID	Req.	What is verified	Method and tool	Pass criterion
SV-01	SR-001	Passwords are stored only as salted PBKDF2 or bcrypt hashes; no password appears in logs or responses.	Register a user, inspect the user table and the application logs; extend TC-AUTH-01 step 5 with a grep of the logs.	Stored value starts with a PBKDF2 or bcrypt identifier; the password text is found nowhere.
SV-02	SR-002	Only TLS 1.2 and above is accepted; HTTP is redirected or refused.	Run testssl.sh (or nmap ssl-enum-ciphers) and curl against the deployed HTTPS build.	TLS 1.0 and 1.1 refused; http:// redirects to https://; no weak cipher reported.
SV-05	SR-005	Users cannot read or change other users' data; every listed action is written to the audit log without raw data or credentials.	Change dataset and run IDs in URLs and API calls as another user; open admin pages as a Student; review the audit log (TC-SEC-01).	403 or 404 and no data returned; log shows the denied attempts with user ID and timestamp.
SV-07	SR-007	Sessions expire, cookies carry the HttpOnly and Secure flags, repeated failed logins are blocked.	Log in, then reuse the session after the (shortened) inactivity limit; inspect cookies in the browser tools; submit 5 wrong passwords in 10 minutes and then the right one.	Expired session is redirected to login; flags present; sixth attempt is blocked for 15 minutes even with the right password.
SV-08	SR-007	State-changing requests without a valid CSRF token are refused.	Replay a POST (upload, train, delete) from Postman without the token and with a wrong token.	HTTP 403 and no state change.

Integrity checks

ID	Req.	What is verified	Method and tool	Pass criterion
SV-03	SR-003	Upload validation rejects wrong type, wrong MIME type, oversized, empty and malformed files; formula cells are neutralised in exports.	Upload the invalid files of TC-DATA-02, then fuzz the upload with truncated, binary and very wide CSV files (Postman or a Python script).	Every invalid file is rejected with a clear message and nothing is stored; no unhandled exception.
SV-04	SR-004	Uploaded model files are refused, only system-created artifacts are loaded, and artifacts cannot be fetched by URL.	Send a .pkl and a .joblib file to /api/upload and any other endpoint; request an artifact path directly; search the code for joblib.load and pickle.load calls.	Model files refused; direct artifact URL returns 404; every load call uses a path created by the system.

Availability checks

ID	Req.	What is verified	Method and tool	Pass criterion
SV-06	SR-006	Request size, request rate, job quota and job time-out limits are enforced.	Send an 11 MB body; send 130 requests in one minute as one user; start a third job as a Data Scientist and a second job as a Student; run a job that exceeds 300 s (lower the limit in the test settings to shorten the run).	HTTP 413; HTTP 429 after the 120th request; extra jobs refused with a message; long job ends as Failed with reason time-out; the server answers other users throughout.

Checks across all objectives

ID	Req.	What is verified	Method and tool	Pass criterion
SV-09	All SR	Automated scan finds no high-severity web vulnerability.	OWASP ZAP baseline scan against the Docker build, run in the CI pipeline or manually before each sprint demo.	No High alerts open; Medium alerts reviewed and recorded.
SV-10	All SR	Static analysis finds no unresolved high-severity security issue in the code.	Run SonarQube (or bandit and flake8) in GitHub Actions on every pull request.	No open blocker or high-severity security finding at merge.

Tools: OWASP ZAP baseline scan, Postman negative tests, testssl.sh or nmap for TLS, SonarQube or bandit in GitHub Actions, and manual abuse cases derived from the STRIDE table in SAD 3.9. A failed security check is a high-severity defect; a failed check that exposes data between users or bypasses authentication is critical.

6. Test Environment

Hardware: Developer laptops (minimum 8 GB RAM, multi-core CPU) for functional tests; one fixed machine for performance tests so results are comparable.

Software: Python 3.11, Django 5, pandas, scikit-learn, joblib, SQLite (development) and optionally PostgreSQL, Docker; browsers: current Chrome and Firefox; operating systems: Windows, Linux, macOS.

Tools: pytest and pytest-cov (unit and integration), Selenium (UI automation), Postman (API), JMeter or Locust (performance), OWASP ZAP (security scan), GitHub Actions (CI), GitHub Issues and Projects (defect tracking).

Test data: Small sample datasets (for example Iris-like and Titanic-like CSV files), a generated 10 MB CSV, a generated 50,000-row CSV, and invalid files (wrong type, empty, oversized, malformed, formula-injection cells). Dummy accounts for each role: Data Scientist, Student and Admin.

7. Test Schedule

The schedule follows the sprint plan in Mini-Project Deliverables Part-2.

Milestone	Dates	Owner
Test case design and review	10-Oct-2026 to 12-Oct-2026	Members 2 and 4
Environment setup, CI pipeline, sprint dry run	12-Oct-2026 to 16-Oct-2026	Members 1 and 3
Sprint 1 test execution	19-Oct-2026 to 23-Oct-2026	All members
Sprint 2 test execution and regression	26-Oct-2026 to 30-Oct-2026	All members
Final UAT and Test Summary Report	29-Oct-2026 to 30-Oct-2026	Member 1

8. Test Deliverables

- Test Plan (this document)
- Test cases (manual and automated), Section 13 and Appendix A
- Test scripts (pytest, Selenium, Postman collection, JMeter or Locust plan)
- Test data sets
- Test execution logs and CI run reports
- Defect reports (GitHub Issues)
- Sprint 1 and Sprint 2 demo videos
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
Test Lead and document owner	Member 1	Introduction, test items, environment, schedule and deliverables; merges and formats the final plan
Test design lead	Member 2	Features, strategy, security validation and SRS traceability
Risk and quality lead	Member 3	Risks, assumptions, suspension criteria, metrics and approvals
Test case author	Member 4	Writes and maintains the 10 test cases (Appendix A) and extends the RTM (Section 13) with design references
Repo owner / reviewer	Team repo owner	Reviews pull requests and confirms tests pass before merge
Faculty guide	Course faculty	Reviews and signs off test results and deviations

10. Risks and Mitigation

Risk	Mitigation
Short two-sprint timeline leaves little time for testing	Write tests alongside features, automate in CI, and test the minimal end-to-end flow first
Delay in delivering a stable build	Merge small pull requests often and run a smoke test on every merge
Different library versions across machines cause false failures	Pin versions in requirements.txt and run tests in Docker and in GitHub Actions
Performance results vary between machines	Run performance tests on one fixed machine and record its specification
Training jobs run long and block test runs	Use small datasets for functional tests and a training timeout
Team members unfamiliar with Selenium or JMeter	Pair on the first script and share templates in the repo
Requirement IDs change after SRS review	Keep IDs in Sections 3 and 13 and in the SRS ID column of Appendix A, and update the three together

11. Assumptions and Dependencies

- Requirement IDs follow SRS Part B and the SAD design references; the plan is updated if the SRS or SAD change after review
- Sample and generated test datasets are prepared before Sprint 1
- Each team member can run the Docker environment locally
- GitHub Actions is enabled on the repository before Sprint 1
- The model catalogue and limits in SRS Tables 4.2-1 and 4.3-1 stay unchanged during the sprints
- A deployed build with HTTPS is available before the security checks in Sprint 2

12. Suspension and Resumption Criteria

Suspend testing if:

- The test environment or CI pipeline is unavailable for more than 4 hours
- The build is too unstable, blocking more than 30% of planned test cases
- A critical defect (data loss, authentication bypass, data exposure between users) is found

Resume testing when:

- Blocking defects are fixed and verified
- The environment is restored and a smoke test passes

13. Test Case Management and Traceability

The Requirements Traceability Matrix (RTM) maps every SRS requirement to the test case or security check that verifies it. The full test cases are in Appendix A (seven functional and three non-functional). The design references come from SAD Appendix A. The SRS RTM (SRS section 8) shows the same mapping from the requirement side and also traces each requirement to its use case and architecture component.

Req ID	Requirement (short)	Design reference	Test case(s) / check	Level	Status
MLE-F-001	User registration and login	DS-Auth-01	TC-AUTH-01	System	Defined
MLE-F-002	Role-based access (Data Scientist, Student, Admin)	DS-Auth-02	TC-SEC-01	System	Defined
MLE-F-003	Upload CSV dataset with validation	DS-Data-01	TC-DATA-01, TC-DATA-02	Integration	Defined
MLE-F-004	Preview and profile the dataset	DS-Data-02	TC-DATA-01	Integration	Defined
MLE-F-005	Handle missing values and encode categories	DS-Data-03	Planned (Sprint 1)	Unit, integration	Planned
MLE-F-006	Configure target column and train/test split	DS-Data-04	TC-MODEL-01	System	Defined
MLE-F-007	Select a model from the catalogue	DS-Model-01	TC-MODEL-01	System	Defined
MLE-F-008	Configure hyperparameters	DS-Model-02	TC-MODEL-01	System	Defined
MLE-F-009	Train a model and show status	DS-Model-03	TC-MODEL-01	System	Defined
MLE-F-010	Compute accuracy, precision, recall, F1	DS-Eval-01	TC-EVAL-01	System	Defined
MLE-F-011	Show confusion matrix and plots	DS-Eval-02	TC-EVAL-01	System	Defined
MLE-F-012	Feature importance / explanation	DS-Eval-03	Planned (Sprint 2)	Unit, system	Planned
MLE-F-013	Save each run (history)	DS-Exp-01	TC-EXP-01	System	Defined

Req ID	Requirement (short)	Design reference	Test case(s) / check	Level	Status
MLE-F-014	Compare runs side by side	DS-Exp-02	TC-EXP-01	System	Defined
MLE-F-015	Export results report	DS-Exp-03	TC-DATA-02 (step 3) (planned, Sprint 2)	System	Partly defined
MLE-F-016	Admin manages users and datasets	DS-Admin-01	TC-SEC-01	System	Defined
MLE-NF-001	Upload of up to 10 MB processed in 5 s or less	SAD 4.3.3	TC-PERF-01	Performance	Defined
MLE-NF-002	Training of standard models on up to 50,000 rows in 60 s or less	SAD 4.2.3, 4.4.3	TC-PERF-02	Performance	Defined
MLE-NF-003	Pages load in 3 s or less for 20 concurrent users	SAD 4.3.1	TC-PERF-03	Performance	Defined
MLE-NF-004	Runs on Windows, Linux and macOS via Docker	SAD App. B	Planned (Sprint 2)	System	Planned
MLE-NF-005	Usable by a first-time user without training	SAD 4.5	Planned (Sprint 2)	Usability	Planned
MLE-NF-006 (new)	Failed jobs are contained; saved data survives restart	SAD 4.4.3	Planned (Sprint 2)	System	Planned
MLE-NF-007 (new)	Keeps page-load limits with 20 datasets and 100 runs per user	SAD 4.3.1	Planned (Sprint 2)	Performance	Planned
MLE-SR-001	Passwords stored hashed	SAD 4.2.1	TC-AUTH-01, SV-01	System	Defined
MLE-SR-002	All traffic over HTTPS (TLS 1.2+)	SAD 4.3.1	SV-02 (planned, Sprint 2)	Security	Planned
MLE-SR-003	Uploads validated for type, size and content	SAD 4.3.3	TC-DATA-02, SV-03	Integration	Defined

Req ID	Requirement (short)	Design reference	Test case(s) / check	Level	Status
MLE-SR-004	Only system-created model artifacts are loaded	SAD 4.1.5 DD3	SV-04 (planned, Sprint 2)	Security	Planned
MLE-SR-005	Per-user data isolation and audit log of actions	SAD 4.4.4	TC-SEC-01, SV-05	System	Defined
MLE-SR-006 (new)	Request, job and run-time limits against abuse	SAD 4.3.1, 4.4.3	TC-MODEL-01 (job limit, step 5); SV-06 (planned, Sprint 2)	Security	Partly defined
MLE-SR-007 (new)	Session expiry, login throttling and CSRF protection	SAD 4.3.7	TC-AUTH-01 (cookie flag only), SV-07, SV-08 (planned, Sprint 2)	Security	Partly defined
MLE-SO-001 to 003	Security objectives (confidentiality, integrity, availability)	SAD 3.9	Covered by the checks for MLE-SR-001 to SR-007 (section 5.1)	Security	Defined

Coverage summary

Requirement group	Total	Fully covered	Open items
Functional (MLE-F)	16	13	F-005, F-012, F-015
Non-functional (MLE-NF)	7	3	NF-004, NF-005, NF-006, NF-007
Security (MLE-SR)	7	3	SR-002, SR-004, SR-006, SR-007

“Fully covered” means a formal test case of Appendix A exists for the whole requirement. Items marked “partly defined” have a test case for one part (the export step for MLE-F-015, the job limit for MLE-SR-006, the cookie flag for MLE-SR-007). Every security requirement also has a security check (SV-nn) in section 5.1. The open items are scheduled in Sprint 1 or Sprint 2 (section 7); this table is updated as their test cases are added to Appendix A.

14. Test Metrics and Reporting

Metrics collected:

- Percentage of test cases executed
- Percentage passed and failed
- Defect density and defect aging
- Requirement coverage from the RTM

- Code coverage from pytest-cov (target 70% or higher for service modules)

Reports:

- Daily execution status during each sprint week
- CI test results on every pull request
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
Test Lead		
Dev Lead		
Faculty guide		

Appendix A: Test Cases

Seven functional test cases (TC-AUTH-01, TC-DATA-01, TC-DATA-02, TC-MODEL-01, TC-EVAL-01, TC-EXP-01, TC-SEC-01) and three non-functional test cases (TC-PERF-01, TC-PERF-02, TC-PERF-03). Author: Member 4. Columns follow the Jackfruit Phase-3 test case template. Actual Result and Test Result are recorded during Sprint 1 and Sprint 2 execution.

Test Case ID	Name of Module	SRS ID	Test Case Description	Preconditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
TC-AUTH-01	Auth and Access Control	MLE-F-001, MLE-SR-001, MLE-SR-007	Login with valid and invalid credentials	A registered Data Scientist account and an Admin account exist.	1. Open the login page. 2. Enter valid username and password, submit. 3. Log out. 4. Enter valid username with a wrong password, submit. 5. Inspect the stored password field in the database. 6. Try to register with a weak password. 7. As Admin, deactivate the account, then try to log in with it.	User: ds_test1; wrong password: Wrong#123; weak password: short1 (fewer than 8 characters)	Step 2: user reaches the dashboard; session cookie is HttpOnly. Step 4: login refused with a generic error, no session created. Step 5: stored value is a salted hash, never plain text. Step 6: registration refused with a message per faulty field. Step 7: login refused for the deactivated account.	To be recorded on execution	Pending
TC-DATA-01	Data Workspace	MLE-F-003, MLE-F-004	Upload a valid CSV and view its profile	User is logged in.	1. Open the Upload page. 2. Select a valid CSV with a header row and submit. 3. Open the dataset preview.	sample_1000rows.csv (about 1 MB, 1,000 rows, 5 columns, some missing values)	File accepted and saved in the user's own directory. First rows previewed. Profile shows row count, column names and types, and missing counts that match the file.	To be recorded on execution	Pending
TC-DATA-02	Data Workspace, Input Validation	MLE-F-003, MLE-F-015, MLE-SR-003	Reject invalid or unsafe uploads	User is logged in.	1. Upload each invalid file in turn. 2. Observe the response. 3. Export a results CSV that contains cells starting with =, +, - or @ and open it. 4. Check the file store.	notes.txt (wrong type); short_19rows.csv (19 data rows); exact_10MB.csv (accepted); empty.csv; oversize_11MB.csv; no_header.csv; formula_cells.csv (cells start with =, +, -, @)	notes.txt: 415 'Only CSV files are supported.' oversize_11MB.csv: 413 'The file is larger than 10 MB.' empty.csv, no_header.csv and short_19rows.csv: 400 VALIDATION_ERROR naming the problem. exact_10MB.csv: accepted. Step 3: formula-like cells are neutralised in the export. No rejected file is stored.	To be recorded on execution	Pending

Test Case ID	Name of Module	SRS ID	Test Case Description	Preconditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
TC-MODEL-01	Model Selection and Training	MLE-F-006 to F-009, MLE-SR-006	Select a model, set parameters and train	A valid dataset is uploaded and prepared.	1. Choose an 80/20 train/test split. 2. Select Logistic Regression. 3. Enter a valid C value, then an out-of-range value. 4. Start training and watch the status. 5. As a Student with one active job, start a second job.	iris_like.csv, target: species; C = 1.0 (valid), C = 101 (above the 0.01 to 100 range); a Student account	Out-of-range value refused; the message names the parameter and range (C: 0.01 to 100). Status moves Queued, Running, Completed. Model artifact stored for this user only; run appears in history. Step 5: second job refused with the limit stated (Student: 1 active job).	To be recorded on execution	Pending
TC-EVAL-01	Evaluation	MLE-F-010, MLE-F-011	Evaluation metrics and confusion matrix	A model has finished training (TC-MODEL-01).	1. Open the results page for the run. 2. Read accuracy, precision, recall and F1. 3. Open the confusion matrix chart. 4. Compute the same metrics with scikit-learn on the same split.	Run from TC-MODEL-01; same random seed in the reference script	Displayed metrics equal reference values within 0.001 (macro-averaged when the target has more than two classes). Confusion matrix renders; cell counts add up to the test set size.	To be recorded on execution	Pending
TC-EXP-01	Experiment Management	MLE-F-013, MLE-F-014	Save and compare experiment runs	Two completed runs exist on the same dataset with different models.	1. Open the experiment history. 2. Confirm both runs are listed. 3. Select both and open Compare. 4. Try to compare one of them with a run on a different dataset.	Runs: Logistic Regression and Random Forest on iris_like.csv	Each run shows dataset, model, parameters, metrics and time. Comparison shows both side by side with values equal to each run's own page. Step 4: mixed selection refused with an explanation.	To be recorded on execution	Pending
TC-SEC-01	Auth, Admin and Audit	MLE-F-002, F-016, MLE-SR-005	Access control and data isolation	User A (Data Scientist) owns a dataset; User B (Student) and an Admin exist.	1. As User B, request User A's dataset and run by changing the ID in the URL and API call. 2. As User B, open an admin page. 3. As Admin, open user management. 4. Review the audit log. 5. As Admin, try to open the contents of User A's dataset.	Dataset ID owned by User A; accounts ds_test1, student_test1, admin_test1	Steps 1 and 2: denied (403 or 404), no data returned. Step 3: Admin page opens. Audit log shows the denied attempts with user ID and timestamp and no raw data values. Step 5: Admin sees dataset metadata only, never the contents.	To be recorded on execution	Pending
TC-PERF-01	Data Workspace	MLE-NF-001	Upload processing time for a 10 MB file	Test machine idle; generated 10 MB CSV available.	1. Upload the file and time it from submit until the profile is displayed. 2. Repeat five times.	generated_10MB.csv	Each of the five runs completes in 5 seconds or less.	To be recorded on execution	Pending

Test Case ID	Name of Module	SRS ID	Test Case Description	Preconditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
TC-PERF-02	Model Training	MLE-NF-002	Training time for 50,000 rows	Generated dataset with 50,000 rows and 20 feature columns uploaded and prepared.	1. Train each of the four catalogue models with default settings. 2. Record time from start to Completed. 3. While training, load the dashboard.	generated_50k_rows_20_features.csv	Training completes in 60 seconds or less for each of the four models. Dashboard stays responsive while training runs.	To be recorded on execution	Pending
TC-PERF-03	API Gateway, Web UI	MLE-NF-003	Page load time with 20 concurrent users	Docker build running; 20 test accounts with sample runs.	1. Configure JMeter or Locust with 20 concurrent users. 2. Each user logs in and opens dashboard, history and results repeatedly for 5 minutes. 3. Collect response times and error counts.	20 dummy accounts; fixed test machine	Page loads complete in 3 seconds or less (average and 95th percentile). No server errors.	To be recorded on execution	Pending